25901191 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCSC202 - INTRODUCTION TO DATABASE MANAGEMENT SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define a field in a DBMS. [K] / [CO1]
2. Explain one-to-one relationship with an example. [U] / [CO2]
3. Differentiate between a single line and a dashed line in an ER diagram. [U] / [CO2]
4. Name the components of a MySQL table. [K] / [CO3]
5. List any four privileges that can be granted in MySQL. [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Explain how DBMS reduces data redundancy. [U] / [CO1]
7. Differentiate between data and information with examples. [U] / [CO1]

8. Explain the role of end users with examples. [K] / [CO1]

9. Describe how referential integrity ensures the connection between two related tables. [U] / [CO2]

10. Describe a scenario where partial dependency and transitive dependency occurs. [U] / [CO2]

11. Explain the difference between COUNT(*), COUNT(column), and COUNT(DISTINCT column) with examples. [U] / [CO3]

12. Explain the UPDATE statement in MySQL. Demonstrate how to update a single column, multiple columns, and update based on a condition with examples. [U] / [CO3]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Explain the relational data model in detail. Describe domain, attribute, tuple, and relation with examples. [U] / [CO2]

14. Using the table EMPLOYEE(EmpID, Name, Department, Salary), write SQL queries to demonstrate: (a) WHERE clause to filter employees with Salary > 30000, (b) ORDER BY to sort employees by Name, (c) GROUP BY to count employees per department, (d) HAVING to show departments with more than 2 employees, (e) LIKE to find employees whose name starts with 'A', (f) BETWEEN to find employees with salary between 20000 and 50000.Explain the purpose of each clause. [A] / [CO3]

[15x1 = 15]